Assignment 7: Custom Hermite-like Interpolator

PV comparison

All PVs use the supplied IFR data, FLAT extrapolation and the value date 2026-01-02. Within each pair only the interpolation method changes, from piecewise constant left-continuous (PCP) to my monotone convex interpolator.

Product	PCP PV	Monotone convex PV	Difference
FRA, May–Nov 2026 (receive fixed 3.2%)	2,908.29	3,259.38	+351.10
RFR swap, Apr 2026–Apr 2029 (receive fixed 3.4%)	6,222.52	12,852.81	+6,630.29
Basis swap, Jul 2026–Jul 2028 (receive SOFR + 10 bp, pay FF)	8,351.35	7,613.82	−737.53

Why they differ

Both IFR curves increase. The left-continuous PCP curve applies the *next* node's value across each whole interval, which is the interval's maximum. My monotone convex curve rises gradually between the node values, so its forwards are lower almost everywhere. Figure 1 shows the gap: zero at each node, widest just after a node, and growing with the size of the jump between nodes. The funding curve references SOFR-1B, so discount factors change too, but that is a second-order effect next to the change in projected floating payments.

- **FRA.** The accrual period (about 0.34–0.84 years) sits on the flat, closely spaced short end, where node values differ by only 10–20 bp. The forward over the period comes out a few bp lower, and the receive-fixed FRA gains about 351.
- **RFR swap.** This is the largest effect. The swap runs to April 2029, through the steep part of the curve. On (2Y, 3Y] PCP uses 3.5% while my curve runs from 3.0% to 3.5%. On the final stub past 3Y, PCP already jumps to the 5Y node value of 4.7%, while my curve is still near 3.5%. The floating leg is much smaller, so the receive-fixed swap gains about 6,630.
- **Basis swap.** Both legs fall, but by different amounts. SOFR's node values rise faster than FF's (for example, +50 bp vs +40 bp from 2Y to 3Y), so moving from step values to the gradual curve takes more off the SOFR forwards than off FF. The basis I receive narrows, and the swap loses about 738.

In short, PCP is biased upward on an increasing curve, and the size of that bias depends on how steep the curve is over each product's dates. A continuous interpolator removes the bias and makes PVs depend smoothly on the dates, with no jumps when a date crosses a node.

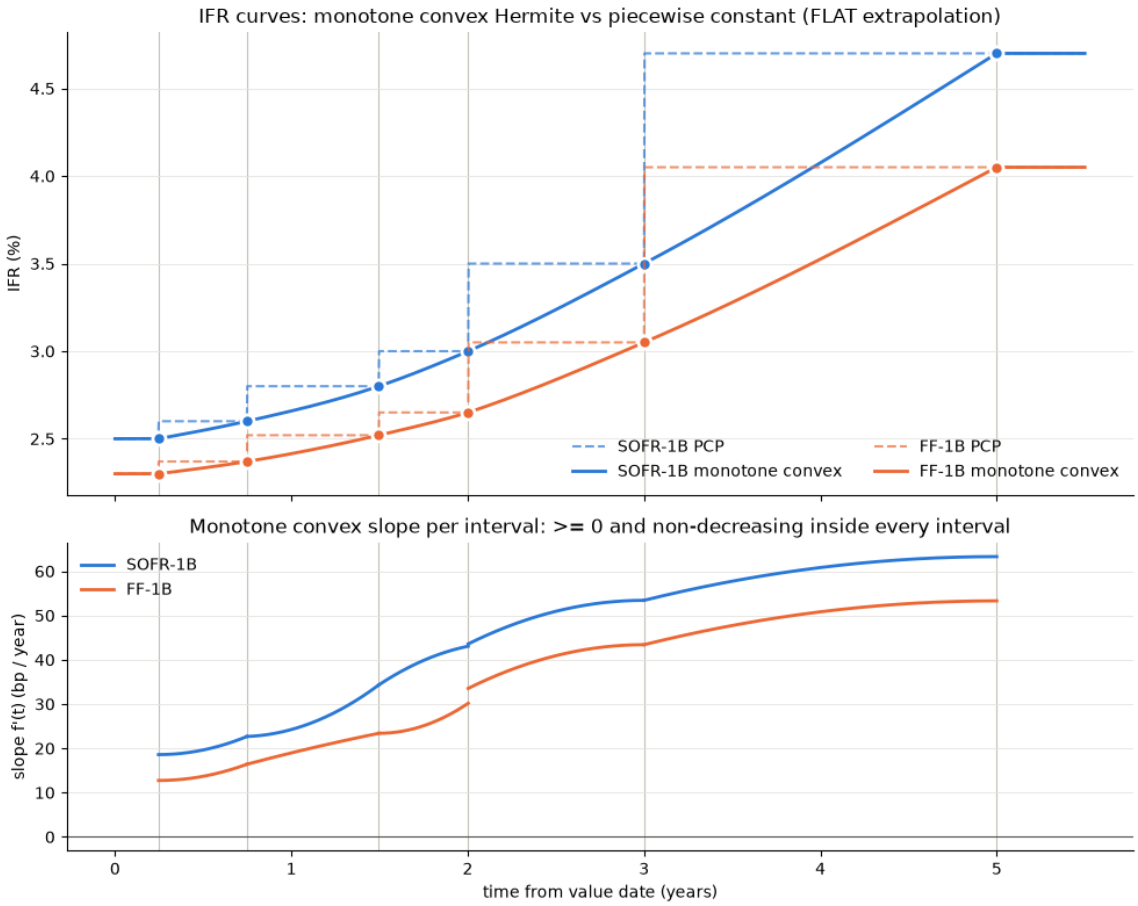


Figure 1: Top: SOFR and FF IFR curves under PCP (dashed) and my monotone convex interpolator (solid), with the input nodes. Bottom: slope $f'(t)$ of my interpolator on each interval; it is non-negative and non-decreasing inside every interval (monotone and convex). The small jump at the 2Y node is the one slope kink the method allows.